PROJECT REPORT
PHISHING URL AWARENESS CHECKER

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Technology: Python 3
Category: Cybersecurity
Project type: Command-Line Application
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1. Abstract

The Phishing URL Awareness Checker is a Python command-line project designed to help users notice common warning signs in a web address. It checks URL text for six indicators associated with suspicious links. It does not open, visit, or submit the URL to a website. It is an educational awareness tool, not a final verdict about whether a website is safe or malicious.

2. Introduction

Phishing is an online deception in which attackers use misleading links to trick people into sharing information or taking unsafe actions. Some URLs contain visible clues, such as an IP address instead of a domain name, an unusual number of hostname labels, or the use of HTTP. This project uses basic Python URL parsing to highlight selected clues.

3. Problem Statement

Users may not notice unusual features in a link before clicking it. A simple tool can inspect the text of a URL and report selected warning indicators without visiting the destination.

4. Objectives

- Develop a lightweight command-line program using Python.
- Check URLs for six specified warning indicators.
- Display clear and understandable warnings.
- Avoid opening or visiting submitted URLs.
- Explain the limitations of basic rule-based checks.

5. Tools and Technologies

Language: Python 3.10 or later
Libraries: Python standard library, including ipaddress and urllib.parse
Interface: Command line
External packages: None required.

6. System Requirements

A computer with Python 3 installed and a terminal or command prompt is sufficient. The project does not require Kali Linux, VirtualBox, or a high-performance computer.

7. Working Methodology

The user enters a URL as text. The program parses the URL, checks the requested features, and displays any indicators found. It does not connect to the URL.

8. Six URL Checks

No.	Indicator	What is checked
1	@ symbol	Whether the URL contains @.
2	URL length	Whether the URL is longer than 100 characters.
3	IP hostname	Whether the hostname is an IPv4 address.
4	Hostname labels	Whether the hostname has 5 or more labels.
5	Punycode	Whether the hostname contains xn--.
6	HTTP scheme	Whether the URL uses HTTP instead of HTTPS.

9. Output and Result Interpretation

If indicators are detected, the tool reports warnings and advises the user to review the link carefully. If no indicator is detected, the URL has only passed these six basic checks. A no-warning result does not prove that a website is genuine or safe.

10. Testing

The project includes sample test conditions for the six indicators. Run the program and verify the results on your own computer before final submission. Capture genuine screenshots of the program and its test results.

11. Advantages

- Easy to use and beginner-friendly.
- Lightweight application.
- No external packages required.
- Does not open suspicious links.
- Helps improve phishing awareness.

12. Limitations

- It checks only six predefined indicators.
- A phishing URL may not contain these indicators.
- A legitimate URL may also trigger warnings.
- It does not check website reputation, page content, certificates, or redirects.
- A no-warning result cannot guarantee safety.

13. Future Improvements

- Add more URL inspection rules.
- Add optional trusted threat-intelligence checks with privacy safeguards.
- Create a simple graphical user interface.
- Generate a detailed URL analysis report.

14. How to Run

1. Install Python 3.10 or later.
2. Extract the ZIP file.
3. Open a terminal in the project folder.
4. Run **python phishing_checker.py**.
5. Enter a URL. Type **q** to quit.

15. Conclusion

The Phishing URL Awareness Checker demonstrates a beginner-friendly use of Python for cybersecurity awareness. It identifies six visible URL indicators and displays warnings in a simple command-line interface. It reinforces the lesson that suspicious links should be reviewed carefully and that the absence of a warning is not proof of safety.

16. Submission Checklist

- Include the Python source code, README, and requirements file.
- Run the program and test cases on your own computer.
- Replace example screenshots with genuine screenshots from your computer.
- Upload the complete project to a public GitHub repository.
- Add the public GitHub repository URL to your final submission.

Important: This report and source package are prepared project materials. Complete local testing and capture your own screenshots before submitting.